PLAYTEST EDITION · BUILD 0.16

FOR FACILITATORS & INSTRUCTORS

COMPANION TO THE FIELD MANUAL

COGNITIVE·TERRAIN

PRACTITIONER'S MANUAL · RUN IT · BUILD IT · KNOW WHY



RUN IT WITH A GROUP

Session plans from a lunch break to a multi-organisation exercise, and how to lead the debrief.

BUILD YOUR OWN CAMPAIGN

The campaign and technology editors, balance checks and sharing.

WHY IT WORKS THIS WAY

Design notes: what each mechanic models, and how it was arrived at.

EDUCATIONAL WARGAME · TURN-BASED
PLAYERS NEED ONLY THE FIELD MANUAL

CT · PM · 2026

FOR WHOEVER RUNS THE ROOM

BEFORE YOU BRIEF

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WHO THIS IS FOR

Instructors, exercise designers and facilitators who run Cognitive Terrain for other people, build campaigns for their own audience, or need to explain why the game models cognitive warfare the way it does.

Players need only the **Field Manual**, which is also built into the game as the Guide (**?**). This manual assumes you have read it or played a campaign.

WHAT PLAYERS SHOULD LEAVE WITH

The target of cognitive warfare is the quality of the other side's decisions, not territory or attention.

Reach is not effect: a viral post on the wrong area changes nothing that matters.

Attack and defence happen at once, and open and closed societies are vulnerable in different ways.

FIRST TIME RUNNING IT?

Play **Typhoon Ines** on **Standard** once yourself, end to end, before you run it for anyone else. It takes about forty minutes.

FOUR FORMATS · SUGGESTED TIMINGS

RUNNING A SESSION

The game runs offline on any computer with a browser, so the format is set by your time and your room, not by the technology. Timings are suggestions; adjust them to your group.

SOLO

ONE PLAYER · 30–45 MIN

1. Send the link or the HTML file with the Field Manual.
2. Ask for **Typhoon Ines** on **Recruit** with the tutorial first.
3. Ask them to bring a screenshot of the debrief and one turning point from the replay.

CLASSROOM

GROUPS OF 2–3 · 50–90 MIN

1. 10 min: the big idea (Field Manual page 3) on the projector.
2. 30–60 min: each group plays one campaign on one screen, one driver, others advising.
3. 15–20 min: debrief as a room (page 5 of this manual).

PAIRED SEMINAR

BLUE vs RED · 60–90 MIN

1. Pair students; one plays Blue, one Red, on two computers in the same room.
2. They swap turn codes by chat (page 6). Sixteen weeks takes about an hour.
3. Debrief each pair, then compare how Red was played across pairs.

DISTRIBUTED EXERCISE

SEVERAL ORGANISATIONS · DAYS

1. Build or choose a campaign and send the file to every cell first (page 7).
2. Cells play one week per day or per session, codes by email.
3. Collect debriefs and replays for a joint hot-wash.

WHAT TO DECIDE BEFOREHAND

Plain or military terms. Plain words suit mixed audiences; military terms (OODA, DQ, Seed Channels) suit staff courses. Players switch in the top bar.

Campaign and difficulty. See page 4.

Week in review on or off. On for learning; off for a faster, harder exercise.

KEEP IT MOVING

Set Red's turn speed to **FAST** once players know what Red's moves look like, and result pop-ups to **KEY ONLY**. Both are in Settings and change nothing in the rules.

EIGHT CAMPAIGNS · THREE DIFFICULTIES

CHOOSING A CAMPAIGN

Every campaign uses the same rules. What changes is the setting, the opponent, the twist at the start and which areas the four crises test. Pick by what you want the room to argue about.

CAMPAIGN	VS	TWIST AT THE START	CRISES TEST	GOOD FOR
Typhoon Ines	Olvana	None	Notice · Interpret · Act · Trust	First games; the core loop
The Valmora Vote	Olvana	Both TRUST areas weakened	Interpret · Notice · Decide · Trust	Elections, polarisation
Collision at Varrin Reef	Olvana	Both officials' entry points weakened	Notice · Act · Act · Decide	Crisis diplomacy, competing videos
Fever in Port Sabre	Olvana	Rumours spread faster for both	Interpret · Trust · Act · Decide	Infodemics, public health
Winter Grid	Donovia	Blue's FOLLOW-THROUGH strained; 4 dormant narratives	Notice · Decide · Act · Trust	Energy coercion
The Lake Referendum	Donovia	Donovia already in Blue's INTERPRET; TRUST weakened	Interpret · Notice · Decide · Trust	Referendums, planted premises
The Silent Survey Ship	Donovia	Both DECIDE areas strained	Notice · Act · Interpret · Decide	Ambiguity, alliance choices
The Vell Valley	Donovia	5 dormant narratives; Blue's NOTICE and TRUST strained	Interpret · Act · Trust · Decide	Annexation and its cover story

DIFFICULTY

LEVEL	WHAT CHANGES
Recruit	Red is slower, makes more mistakes and earns 2 fewer points a week. For first games and short sessions.
Standard	A steady, sensible Red. Campaigns are balanced at this level.
Veteran	Red plans two moves ahead. No extra budget, no hidden information. For experienced players and competitions.

MATCH THE OPPONENT TO THE LESSON

Olvana teaches patience against nerve: a closed state that works one area at a time from inside.

Donovia teaches that the aim can be doubt and inaction rather than persuasion: floods, planted premises, shows of force.

Field Manual pages 24 and 25 set out each one's rules and sources.

SAME MAP, FAIR COMPARISON

Groups playing the same campaign get different maps unless they start from the same seed. For a like-for-like comparison, run it as a two-player game from one Blue code (page 6).

THREE VERDICTS · THE REPLAY · QUESTIONS

LEADING THE DEBRIEF

The game ends with a neutral debrief and a replay of both minds week by week. Most of the learning happens here, so leave time for it.

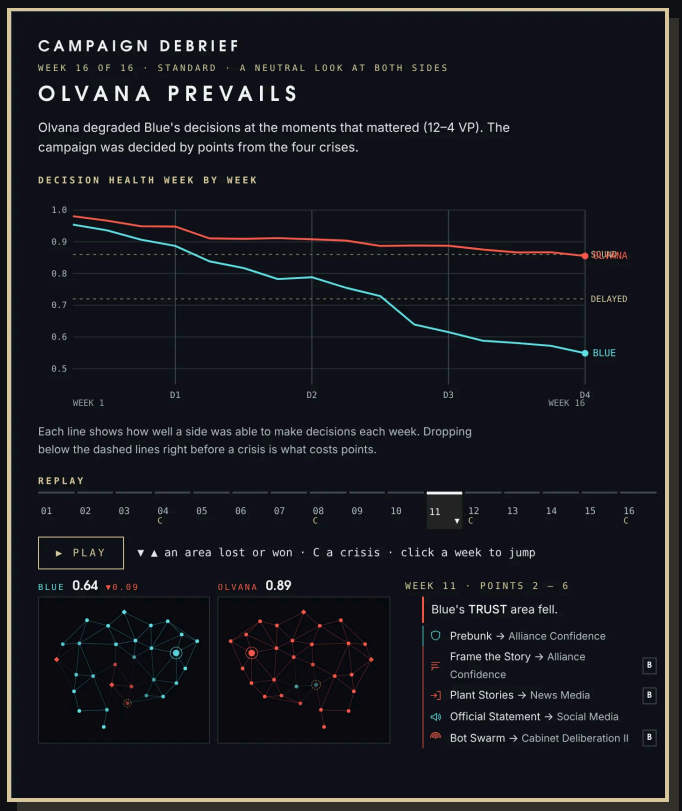


FIG. 1 The debrief: crisis points, health week by week, three verdicts, turning points and lessons.

THE THREE VERDICTS

VERDICT	MEASURES
Who decided better	Decision Health and crisis results across the campaign.
Who adapted more	How many of the eight capabilities each side used, and how many adaptive or protective moves.
Who got more for less	Influence moved per point spent attacking.

A side can win on points and lose two of the three verdicts. That gap is often the best conversation in the room.



FIG. 2 The replay: step through every week, with both minds and that week's moves.

QUESTIONS THAT WORK

- Which crisis did you lose, and when did you first know it was coming?
- Find a move with a great spread score that changed nothing. Why?
- When did defending your own mind matter more than attacking theirs?
- What did Red seem to be trying to do? What told you?
- Which of Blue's advantages as an open society did you use, and which did you lose?
- What would you tell a decision-maker in the real version of this crisis?

TWO PLAYERS · NO SERVER · NO SHARED NETWORK

ACROSS ORGANISATIONS

Two-player games need nothing but a way to send a line of text. That makes the game usable between cells, units or agencies that cannot share a network or install software.

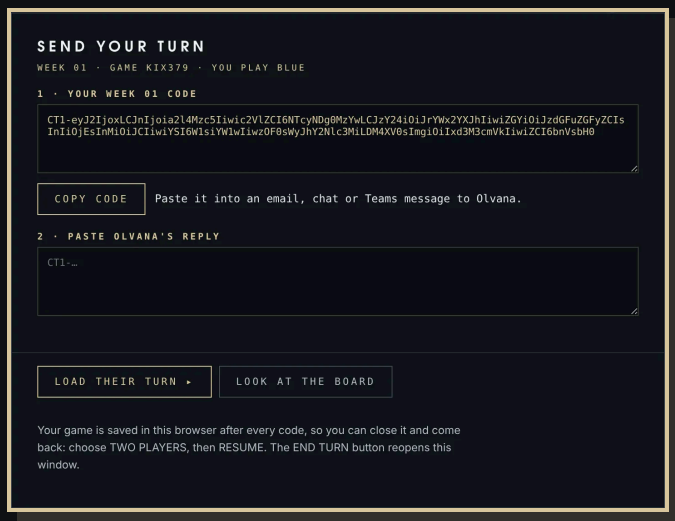


FIG. 3 A turn code with a COPY button, and a box for the reply.

WHAT A CODE CARRIES

The game and its map seed, the campaign, the week, the side and that side's moves. Each copy replays the other's moves and checks the board matches; if it ever does not, the log says so. Codes contain no personal data.

CHECKLIST

1. **Same version.** Codes work only between builds with the same minor version (0.16.x with 0.16.x). Send everyone the same download.
2. **Same campaign.** For a custom campaign, every Red player imports the file before the first code (page 7). The code checks it.
3. **Assign sides.** Blue starts the game; Red joins with Blue's first code. The campaign decides whether Red is Olvana or Donovia.
4. **Set a rhythm.** One week per session or per day keeps a sixteen-week campaign moving.
5. **Collect the debriefs.** Each side sees the same debrief at the end; Red sends one final code so Blue's game ends too.

ANY CHANNEL WORKS

Email, chat, Teams, a text file on removable media. Extra text around a pasted code is ignored, so players can add a note.

Each player's game saves after every code and appears under **RESUME**, so a game can pause for days.

CAMPAIGN EDITOR · SHARE · IMPORT

MAKE A CAMPAIGN

Set the scene for your own audience, place the crises and new technologies on the calendar, tune how strong each side starts, then test it, check its balance and share it. No code involved.

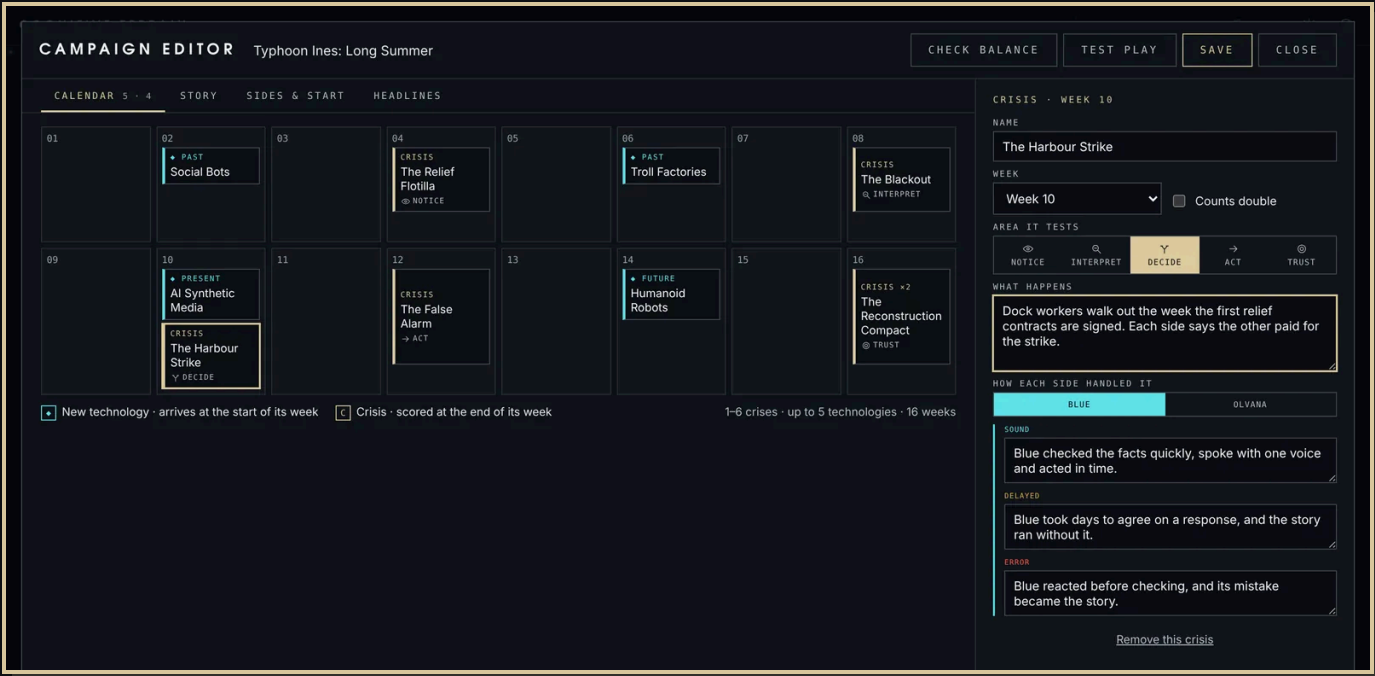


FIG. 4 The campaign editor, editing a copy of Typhoon Ines with a new crisis in week 10. Click a crisis or technology on the calendar to edit it on the right; + adds one in that week.

MAKE ONE

1. On the start screen press **CAMPAIGNS** (or ⚙ → **CAMPAIGN EDITOR**).
2. **+ NEW CAMPAIGN**, or **COPY & EDIT** a built-in one. Copying is the fastest way to a balanced start.
3. Choose the opponent (Olvana or Donovia), write the setting and the headlines' places and outlets, and set any twist at the start.
4. Place 1 to 6 crises and up to 5 new technologies (page 8).
5. **CHECK BALANCE**, then **TEST PLAY**.

SHARING

Save a **.campaign.json** file or copy a code that starts `CTC1-`. Campaigns are plain text and cannot run code, so they are safe to pass around. Campaigns live in the browser that made them: keep the file.



FIG. 5 The campaign library: play, copy, save as a file, copy a share code, import.

WRITING CRISES · CHOOSING TECHNOLOGIES · BALANCE

CRISES & TECHNOLOGIES

WRITING A CRISIS

A crisis is a moment that tests one area of both minds. For each one you write what happens, choose the area it tests, and write how each side handles it at each result: SOUND, DELAYED and ERROR. Players read your text at the moment of truth, so keep each line to one or two sentences about a decision, not an outcome.

> **Pick the area by the decision.** A flood of rumours tests NOTICE; a contested video tests INTERPRET; a deadline tests DECIDE; a delivery that must happen tests ACT; a scandal tests TRUST.

The last crisis counts double. Spread the tested areas so no single defence wins the campaign.

CHECKING BALANCE

CHECK BALANCE plays your campaign 40 times, computer against computer, and reports how often each side wins. The built-in campaigns are tuned so Blue wins roughly 40–60% of games. If yours is far outside that, soften the twist or move a crisis to an area the stronger side has not prepared.

ORIGINAL FICTION

The built-in campaigns invent every place, outlet, person and event. Doing the same keeps a session about methods rather than about a real event, and keeps it usable in any room.

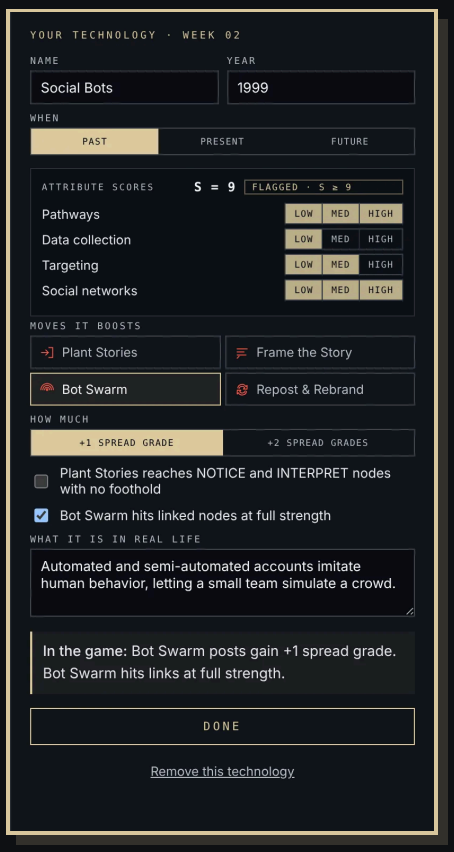


FIG. 6 Building a technology: name, year, era, the four attribute scores and their total S, and which moves it boosts.

TECHNOLOGIES

Pick from 18 real technologies, past, present and future, or **BUILD YOUR OWN**. Each is scored 1 to 3 on four attributes from Rushing, Umeokolo & Xu (2025): new pathways, data collection, targeting and social networks. A total of 9 or more is the paper's mark of a disruptive innovation. Where the paper gives no date or score, the editor marks the figure as the designer's estimate.

OFFLINE · PROJECTORS · SOUND · PRIVACY

SETTING UP THE ROOM

GETTING IT ONTO MACHINES

OPTION	BEST WHEN
Play in the browser (playtest page)	Internet is available and nothing can be installed.
One HTML file	Locked-down or offline machines: copy it over and double-click.
Installers (Windows, Mac, Linux)	Personal machines. Not yet code-signed: Windows needs More info → Run anyway; Mac needs right-click → Open the first time.

ON A PROJECTOR

Settings → **Chart Room** (the light scheme) reads best on washed-out projectors. Set **Interface size** to LARGER or LARGEST for the back row, and turn on **Colour-safe mode** so the sides differ by shape as well as colour.

PRIVACY

The game has no accounts, no analytics and no server; its only network request is for web fonts, and it works without them. Settings, saved two-player games and custom campaigns stay in the browser on that machine. Clearing the browser's site data removes them.

IMMERSIVE BRIEFINGS IN A ROOM

The optional voice pack reads each week in review aloud as the broadcaster of the side ahead (Field Manual page 20). In a room it works best on the presenter's machine with speakers.

1. Download **CognitiveTerrain-Voice-Emma.zip** (about 117 MB) once, before the session.
2. Web or HTML file: Settings → Immersive Briefings → **LOAD VOICE PACK...**. Installers: put the zip beside the game before starting it.
3. Press **HEAR A SAMPLE** to check the speakers.

It runs offline, on that machine only. The voice is synthetic speech from an open-source model; tell your audience so. Slower machines take a few seconds before the first line.

ACCESSIBILITY CHECKLIST

Text size and interface size are separate settings. Every colour scheme passes 4.5:1 contrast. Reduced motion stops pulsing and drifting. Number keys pick moves; Esc cancels. Captions show whenever the narrator speaks.

WHY IT WORKS THIS WAY

The next five pages take each mechanic in turn: what it models, how I arrived at it, and where it comes from. Where I chose playable over precise, I say so.

THE MIND AS TERRAIN

MODELS A nation's decision-making as the ground being fought over.

Cognitive warfare is hard to teach because you can't see it, and most people picture it as "fake news". I wanted the real idea on the table: the target is whether the other side can still make good decisions under pressure. So the board is two brains, each about thirty connected nodes, and every node is a real kind of person or institution: a newsroom, an analysts' cell, a cabinet, a crisis team, the public's trust in its own government.

RUSHING, HERSCH & XU (2026), COGNITIVE WARFARE FRAMEWORK

FIVE AREAS: THE DECISION LOOP PLUS TRUST

MODELS Observe, orient, decide and act, with trust as the slow layer underneath.

The nodes sit in NOTICE, INTERPRET, DECIDE and ACT, after the research's multi-horizon cognitive OODA loop, with TRUST underneath them all. I wanted losing an area to hurt the way losing that capability would, so each loss has one plain effect: you can't see the other mind clearly (NOTICE), every move costs 1 more (INTERPRET), you get one move fewer (DECIDE, also lost if Top Leadership falls), a quarter of your moves misfire (ACT), or nobody believes your statements (TRUST). An area falls when the other side holds half its nodes, a threshold simple enough to read off the map.

RUSHING, HERSCH & XU (2026) · EFFECTS ARE DESIGN ABSTRACTIONS

DECISION HEALTH IS THE SCOREBOARD, NOT REACH

MODELS How well a side can still decide, 0 to 1.

This is the lesson I most want players to leave with: spreading far is not the same as changing minds. So the score is not views or nodes held. It is the weighted health of the four loop areas, multiplied by trust (from 0.75 with trust gone to 1 with it whole), minus 0.06 for each area lost and any flooding penalty. The weights differ by side: a centralized state leans 40% on DECIDE, Blue 30%, because when everything runs through the top, the top is the biggest target. Difficulty never changes the formula.

RUSHING, HERSCH & XU (2026) · WEIGHTS: DESIGNER'S READING OF SHIN (NIU)

ENTRY POINTS AND FOOTHOLDS

MODELS Messages need a way in before they can travel.

Each brain has three entry points (News Media, Social Media, Government Contacts). Attacks start there and spread along links from nodes you have a foothold in. It makes players plan an approach instead of hitting the most valuable node first, which is how influence work actually has to be sequenced.

DESIGN ABSTRACTION

MOVES & METRICS

EIGHT MOVES FROM EIGHT CAPABILITIES

MODELS The attacker's and defender's capability families.

I did not invent a list of tricks. The four attack cards (Plant Stories, Frame the Story, Bot Swarm, Repost & Rebrand) and the four defence cards (Watch Team, Prebunk, Official Statement, Lessons Learned) are the four attacker and four defender capability families in the research, one card each. In military mode they keep doctrinal names. Donovia adds a ninth, Show of Force, because its doctrine pairs stories with visible military pressure.

RUSHING, HERSCH & XU (2026) §2.2.2, §2.3.2 · SHOW OF FORCE: ISW, IISS

THE SPREAD SCORE

MODELS How engaging a campaign of posts actually was.

Every attack rolls an audience and grades it with the research's engagement metric: 0.1 per view, 0.3 per like, 0.7 per comment and 1 per share, divided by the number of posts, on a grade scale from F to A+. I kept the formula and the scale because they teach that interaction outweighs views. The grade then multiplies the move's push (×0.25 for F to ×1.6 for A+); those multipliers are mine. A grade only multiplies the push: a viral post on the wrong area still loses the crisis.

RUSHING & XU (2025) METRIC AND GRADES · MULTIPLIERS: DESIGNER'S

FAILURE AND BACKFIRE

MODELS Platforms, watchers and prepared audiences.

Any attack can be taken down or fall flat (8%). A Watch Team on the target adds 15% that it is exposed; a prebunked audience adds 8% that it shrugs it off. A failed attack nudges the node back toward the defender. I wanted defence to change odds, not just add hit points, so players feel why detection and inoculation matter before a campaign lands.

DESIGN ASSUMPTION

RECOVERY, FRAMING AND SPREAD

MODELS Minds drifting back, and ideas settling in.

Every week each node recovers a little toward its own side, faster with Lessons Learned; strongly held nodes push on their neighbours. Framing stops a node recovering for two weeks, so the next push sticks: it is the game's version of priming. These are design assumptions, tuned so that a single hit fades but a sustained, sequenced campaign holds.

DESIGN ASSUMPTION

THE MOMENTS THAT COUNT

SIXTEEN WEEKS, FOUR CRISES

MODELS Decisions under pressure at the moments that matter.

A campaign is sixteen weeks so it fits a lunch break. In weeks 4, 8, 12 and 16 a crisis tests one area of both minds; the last counts double. Each side's crisis score is 0.6 × its Decision Health plus 0.4 × the health of the tested area, less 0.08 if that area is lost. SOUND (0.86 or more) earns 3 points, DELAYED (0.72 or more) 1, ERROR none. Weighting the tested area makes players read the calendar and prepare, which is the skill I wanted to reward.

DESIGN · OODA FRAMING FROM RUSHING, HERSCH & XU (2026)

COGNITIVE SUPERIORITY

MODELS One side's decision-making clearly dominating the other's.

If one side has lost at least two more areas than the other for two weeks in a row, the campaign ends early. The research defines cognitive superiority as a sustained advantage in the decision loop; the knockout gives that idea a hard edge, and the race toward it shows in the side panel as LEAD.

RUSHING, HERSCH & XU (2026), COGNITIVE SUPERIORITY · THRESHOLDS: DESIGNER'S

NEW TECHNOLOGIES

MODELS Disruptive innovations arriving mid-campaign.

In weeks 2, 6, 10 and 14 a new technology arrives, and both sides choose in secret: use it for an attack bonus, or prepare and deny the other side's. Using it against a prepared opponent gains nothing. I built it as a dilemma because that is the real problem with new tools: you don't know whether the other side is ready. The technologies, their attribute scoring, the scale and the threshold of 9 come from the research, whose forecast predicts a disruptive innovation roughly every 4.6 years. Where the paper gives no score, the figure is marked as my estimate.

RUSHING, UMEOKOLO & XU (2025)

BUDGET AND MOVES

MODELS Finite attention and resources.

Three moves a week from a budget of 6 points plus 1 per enemy node held, capped at 18, with one extra move for points left over. The cap stops hoarding; the extra move rewards spending. Holding the other side's Government Contacts pays 2 more a week, because access to officials is worth more than any post.

DESIGN ASSUMPTION

RED IS NEVER A MIRROR

OPEN SOCIETY AGAINST CLOSED STATE

MODELS Why open and closed societies are vulnerable in different ways.

Blue is an open democracy. Its posts into a censored internet spread less, while Red's posts into Blue's open internet spread further. In return Blue has a free press and courts (its TRUST recovers faster, and Official Statements hit harder while trust holds), truthful messages that stick (won-over nodes drift back at about a third of normal speed) and a cheap extra move. The asymmetry is the point: you fight a closed system differently from how it fights you.

ATP 3-13.1 · SHIN (NIU) · SEVERAL EFFECTS ARE DESIGN ASSUMPTIONS, LABELLED IN THE FIELD MANUAL

TWO OPPONENTS FROM THE ARMY'S TRAINING WORLD

MODELS Two doctrines of cognitive warfare, kept fictional.

I took Red's countries from the Army's Decisive Action Training Environment, because professional wargamers already know them and they keep the game clearly fictional. Both share a core: already inside Blue's social media, run from the top, slow to learn, expensive to improvise. Olvana should feel like a fortress with one gate: patient pressure on one area at a time. Donovia should feel loud: floods, planted premises, dormant narratives going off like mines, and threats that can blow back. Every rule names its source or says it is my assumption, and every real-world claim was checked against the sources in 0.14.2.

ODIN DATE · OLVANA: SHIN (NIU), EADS ET AL. · DONOVIA: NATO C2COE, ISW, IISS, CARNEGIE

A COMPUTER OPPONENT THAT PLAYS FAIR

MODELS A competent adversary, at three levels of skill.

On Recruit the opponent is noisier and earns 2 fewer points a week. On Standard it plays a steady, sensible campaign. I wanted Veteran very hard to beat without cheating, so it plans two moves ahead instead: it tries its best options on a scratch copy of the game, lets the week play out, and keeps the move that leaves it best placed. It gets no extra budget and no hidden information, and its trial runs never touch the game's own dice. The opponent also explains itself: each of its moves shows the most likely reason, judged only from what the player could have seen.

DESIGN

BUILT TO BE USED

TEACHING THROUGH PLAY

The game should explain itself while you play, not ask you to read first. So it explains at every moment that matters: before you commit (the hover card's three numbers), when it happens (a headline and a full story), while the opponent moves (each move lit and explained), after each week (the week in review and Looking ahead) and at the end (three verdicts and a replay). The Guide is the Field Manual itself, built in.

TWO PLAYERS BY TURN CODES

No server, no accounts, nothing to host or break. A short code sent by email or chat works in a classroom and across organisations that can't share a network, which is exactly where this game is most useful.

CAMPAIGNS AS DATA

A campaign is plain text that can never run code, so facilitators can build one for their own audience and share it safely without touching the program. The stories are always invented; the research shapes how each side operates, never what happens.

BALANCE

A campaign is balanced when the computer playing Blue wins roughly 40 to 60% of games against the computer playing Red. Every built-in campaign is checked that way before release, and the editor runs the same check. Twists make campaigns feel different without changing the core rules. The opponent should punish a player who only attacks or only defends, and lose to one who reads the next crisis and prepares.

IMMERSIVE BRIEFINGS

Every week happens twice: once on the board, and once in how each side tells it. The optional voice pack lets players hear the second version, from whichever side is winning, built from the moves they just watched. It is optional because the base game must stay small, quiet and offline.

THE LOOK, AND ACCESS FOR EVERYONE

The brains are drawn like a space game's galaxy map, in graphite and sand, so an abstract subject feels like a place you can hold, and never like a generic interface. Accessibility is a requirement, not polish: one offline HTML file, an interface that scales instead of shrinking, 4.5:1 contrast in every scheme, colour never the only signal, plain words first.

ONE RULE ABOVE THE REST

Keep the original game. New features are built around the core loop, never instead of it.

WHERE THIS COMES FROM

The game is built on published research. The rules simplify it; the debrief and the in-game guide point back to it.

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ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

Changes made: the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; three of the five innovation score sets are the designer's own estimates; all scenario, narrative and tutorial text is original.

No endorsement: the authors and their institutions have not reviewed or endorsed this game.

Immersive Briefings voice pack (optional, separate download): Kokoro-82M by hexgrad, kokoro-js and transformers.js (Apache 2.0); ONNX Runtime (MIT); eSpeak NG (GPL-3.0-or-later), whose source is published with the pack.

HOW THIS GAME WAS MADE

Design and direction: Kevin Williamson. Cognitive Terrain is my design: I conceived the game, built its rules from the research, chose every source, defined both opponents and the campaign briefs, set the balance targets and accessibility standards, and reviewed every rule, campaign and page.

Tools: I used AI-assisted development tools to help implement the code, draft text and run balance tests, working from my specifications; I reviewed, revised or rejected everything they produced. The published sources, not the tools, are the authority for any rule that models real-world behaviour. The art, icons and music are drawn and synthesized by the game's own code; no AI image or music generators were used, and the base game runs no AI while you play. The optional Immersive Briefings voice is synthetic speech from an open-source model that runs on your own computer (Field Manual page 20).

Olvana and Donovia, and the countries Bothnia, Framland, Arnland and Gorgas, are fictional training countries from the US Army's ODIN DATE environment; they are used here without endorsement. All eight scenarios, their towns, outlets, people and every event in them are original fiction. Any resemblance to real places or events is coincidental. Donovia's rules follow the NATO, ISW, IISS and Carnegie research above; game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.